

Chapter 29: Operative Vaginal Delivery — Revision Notes

Williams Obstetrics 26e, pp. 1365–1404 of the PDF. High-yield numbers are in **bold**. The three tables (29-1 to 29-3) are transcribed from the PDF images (they are missing from the extracted chapter text). 13 of the 15 figures are included (Figs 29-4 and 29-5, blade-insertion close-ups, are omitted).

Big picture: Operative vaginal delivery (OVD) = forceps or vacuum traction that adds to maternal pushing. **Station and rotation** decide the risk. The right comparator is **second-stage cesarean**, not spontaneous birth. **Forceps → more maternal (OASIS) injury; vacuum → more scalp/extracranial fetal injury.**

1. Incidence and Indications

- US OVD rate: **3.1% (2015)**, down from **9% (1990)**. **Vacuum : forceps ≈ 4 : 1.**
- **Failure to achieve vaginal birth:**
 - All US (2006): forceps **0.4%**, vacuum **0.8%**
 - Term nulliparas (25 academic hospitals): forceps **4.4%**, vacuum **6.4%**
- **Indication:** any condition threatening mother or fetus that delivery will relieve, if technically safe.
 - **Maternal (most common): maternal exhaustion, prolonged second stage** (no defined maximum second-stage length mandates OVD). Also conditions limiting pushing or needing expedited delivery: severe/acute pulmonary compromise, intrapartum infection with decompensation, neurological disease, serious cardiac disease.
 - **Fetal: nonreassuring FHR, premature placental separation.**

2. Classification and Prerequisites

- **Station** = cm above/below the **interspinous line (0 station)**; range **–5 to 0 to +5**.
- Categories: **outlet, low, midpelvic**; most are low or outlet.
- **⚠ High forceps (above 0 station) have no place in current obstetrics.**

Table 29-1. Operative Vaginal Delivery Prerequisites and Classification According to Station and Rotation^a

Procedure	Criteria
Outlet forceps	Scalp is visible at the introitus without separating the labia ; fetal skull has reached pelvic floor ; fetal head is at or on perineum; head is OA or OP or head is right or left OA or OP position but rotation ≤45 degrees
Low forceps (2 types)	Fetal station is ≥+2 cm but not on the pelvic floor , and: (a) rotation ≤45 degrees is required or (b) rotation >45 degrees is required
Midforceps	Fetal station is between 0 and +2 cm

Prerequisites

Engaged head	Experienced operator	No fetal coagulopathy
Vertex presentation ^{b, c}	Ruptured membranes	No fetal demineralization disorder
Known fetal head position	Completely dilated cervix	Informed consent completed
CPD not suspected	Adequate anesthesia	Willingness to abandon OVD
Fetal weight estimated	Emptied maternal bladder	

^aClassification for the vacuum delivery is the same as for forceps except that vacuum is used for traction but not rotation. ^bForceps, but not vacuum extractor, may be used for delivery of a mentum anterior face presentation. ^cPiper forceps may be used to deliver the head during breech delivery. CPD = cephalopelvic disproportion; OA = occiput anterior; OP = occiput posterior; OVD = operative vaginal delivery.

- SMFM publishes a preprocedural **checklist and documentation template**.
- **Vacuum: ideally not <34 wk** (cranial hemorrhage risk). No recent **fetal scalp blood sampling** before vacuum.
- **Know the head position**; sonography (orbits, nasal bridge) helps when unclear.
- **Analgesia**: regional preferred for low/mid; **pudendal** may suffice for outlet. Newer/early epidurals do **not** ↑ OVD rates.
- **Empty the bladder**. Urinary retention is common after OVD (episiotomy and epidural add risk) → usually resolves with **24–48 h** of catheter drainage.

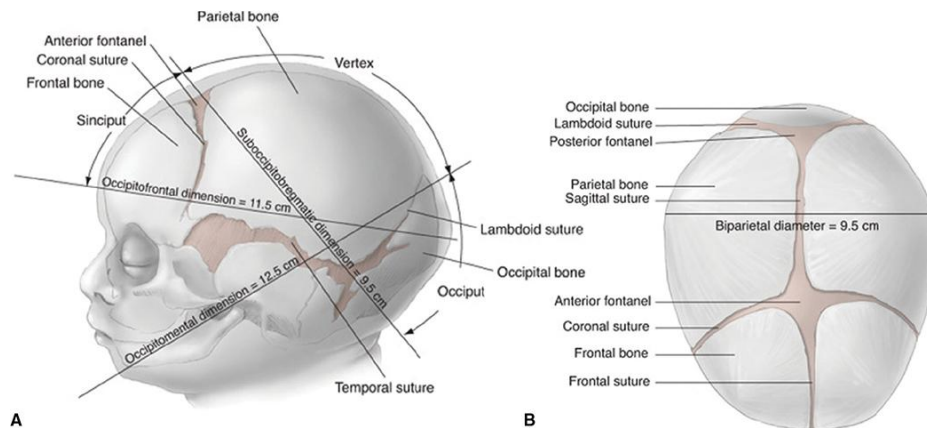


Figure 29-1. Term fetal head: occipitofrontal 11.5 cm, occipitomental 12.5 cm, suboccipitobregmatic 9.5 cm, biparietal 9.5 cm; fontanels and sutures used to confirm position.

3. Morbidity

- Higher station and more rotation → harder procedure, more injury.
- **Compare with cesarean** (the real alternative), not spontaneous birth.

Maternal morbidity

Lacerations

- OVD ↑ **3rd/4th-degree (OASIS)**, vaginal wall and (less often) cervical tears vs spontaneous birth, and far more than cesarean.
- **Forceps > vacuum** for OASIS, especially with a **midline episiotomy**.
- **OASIS protection**:
 - **Indicated, not routine, episiotomy**; if needed, **mediolateral** (better OASIS protection; pain/dyspareunia similar or ↑)
 - **Early forceps disarticulation/removal** (avoid full distention by blades + head)
 - **Stop maternal pushing** during disarticulation/crowning → operator controls the force
 - **Dedicated assistant bolstering the perineum** (authors' practice)
 - **Rotate OP → OA** (manual or forceps), since OP ↑ OASIS

Infection and future pregnancy

- Cesarean, not OVD, is a risk for **peripartum hysterectomy** (>1 million births).
- Avoiding cesarean avoids future previa, accreta, rupture and adhesions. Wound/uterine infection more frequent after cesarean than OVD.
- **ANODE trial** (~3500 OVDs): single IV **amoxicillin–clavulanic acid** → puerperal infection **11% vs 19%** (placebo).

- **ACOG:** prophylaxis "reasonable" but **no routine antibiotics for all OVD**. **Single dose reasonable with 3rd/4th-degree lacerations.**

Pelvic floor disorders

Disorder	Link with OVD
Urinary incontinence	Parity and vaginal birth are risks; OVD not clearly worse than vaginal birth alone
Anal incontinence	Conflicting; OASIS rather than delivery mode is probably the main link (OVD ↑ OASIS). Cesarean not protective long term
Pelvic organ prolapse	Mixed. Levator ani avulsion (esp. with forceps) implicated; OVD without avulsion → no ↑ POP

Perinatal morbidity

- OVD has **higher fetal injury risk than cesarean or spontaneous birth**; risk rises with duration, number of pulls and **pop-offs**.

Instrument	Characteristic injuries
Vacuum (more injuries overall)	Cephalohematoma, subgaleal hemorrhage, retinal hemorrhage , jaundice from these, clavicular fracture, scalp lacerations
Forceps	Facial nerve injury, brachial plexus injury, depressed skull fracture, corneal abrasion
Both	Shoulder dystocia

- **Intracranial hemorrhage:** vacuum higher in some studies, equal in others. Rates are **similar for vacuum, forceps and cesarean during labor**, but higher than spontaneous birth or **prelabor cesarean** → the common factor is **abnormal labor**.
- Werner (>150,000): forceps had **fewer total neurological complications** (seizures, IVH, SDH) than vacuum or any cesarean.
- **Acidemia/HIE** not higher with OVD vs second-stage cesarean.
- **Rotational OVD (Kielland, rotational vacuum) vs second-stage cesarean:** similar maternal and neonatal morbidity.
- **Low vs midforceps:** comparable neonatal morbidity (mostly older data).
- **Midpelvic OVD vs second-stage cesarean (Muraca):**
 - **Dystocia indication:** fetal and maternal morbidity **higher with OVD**
 - **Distress indication:** neonatal morbidity ↑ only with **midpelvic vacuum**; maternal morbidity ↑ only with **midforceps**

Mechanisms

- Cephalohematoma/subgaleal: **suction ± rotation shears vessels**.
- Intracranial hemorrhage: fracture lacerating vessels, or force alone.
- **Facial palsy:** blade compresses the nerve against the facial bones.
- **Brachial plexus:** "**shoulder dystocia at the pelvic inlet**": head descends while shoulders stay above the inlet; traction stretches the plexus.

Long-term: reassuring. No link with **epilepsy** (>21,000) or **cerebral palsy**; neurodevelopment similar after successful forceps, failed forceps + cesarean, or cesarean alone; midpelvic OVD not harmful.

Trial of OVD and sequential instruments

- An expected-difficult OVD is a **trial**, ideally **in the operating room** with immediate cesarean available.
 - Forceps won't apply satisfactorily → **stop** → vacuum trial or cesarean. Vacuum with **no descent** → cesarean.
 - Cesarean after a prepared trial: **no worse neonatal outcome** (reassuring FHR). **Unexpected** failure without preparation → ↑ neonatal morbidity.
- **Failure risks: persistent OP, birthweight >4000 g** (but prediction is poor). Attempt only if success is likely.

- **Sequential instruments** (usually vacuum → forceps) **significantly ↑ fetal trauma** → **ACOG: avoid unless a "compelling and justifiable reason."**

4. Forceps Delivery

Design

- **Forceps** = the pair; each half = a **branch**, named **left/right by the maternal side** it is applied to.
- **Four components: blade, shank, lock, handle.** Blade has a toe, heel and two curves:
 - **Cephalic curve** (outward) fits the fetal head
 - **Pelvic curve** (upward) fits the birth canal

Feature	Options	Trade-offs
Blade	Solid ; fenestrated (true window); pseudofenestrated (smooth outside, indented inside)	Fenestration ↓ head slippage in rotation but ↑ thickness/friction; pseudofenestration aims for both grip and easy insertion
Shanks	Parallel (e.g. Simpson) vs overlapping (e.g. Luikart)	Parallel: ↓ head compression but wider at the introitus. Overlapping: ↑ compression, ↓ perineal distention (best for outlet)
Lock	English (end of shank near handles), pivot (end of handles), sliding (along the shank)	Sliding lock allows correction of asynclitism
Handle	—	Squeezing ↑ compression ; forces = traction + compression

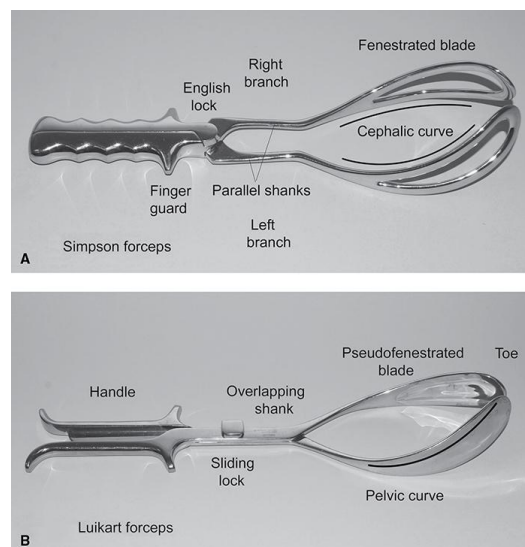


Figure 29-2. Simpson forceps (fenestrated blades, parallel shanks, English lock) vs Luikart forceps (pseudofenestrated, overlapping shanks, sliding lock).

Blade application (OA)

1. **Left branch first:** handle in the **left hand**; fingers of the **right hand** in the left posterior vagina as a guide; blade enters along the **left posterior vaginal wall** and is swept up and in between head and fingers.
2. **Right branch:** right hand holds it; left-hand fingers guide in the right posterior vagina.
3. **Insertion force comes mainly from the thumb** behind the heel.

Oblique positions

- **LOA/ROA:** place the **lower (posterior) blade first**.



Figure 29-3. OA/LOA: the left branch, held in the left hand, is introduced on the left between head and the operator's right-hand fingers.

Checking application

- Ideal grasp: blade long axis along the **occipitomental diameter**; most of the blade over the **lateral face**.
- **OA**: blades **equidistant from the sagittal suture**, each equidistant from its **lambdoid suture**.
- **OP**: symmetric to the sagittal suture and each **coronal suture**.
- **Won't articulate** → reassess suture relationships. Brow + occiput placement → cannot lock, or slips with traction.
- Slow articulation corrects mild **asynclitism**; for greater asynclitism use **Luikart** (sliding lock: slide branches until finger guards align).

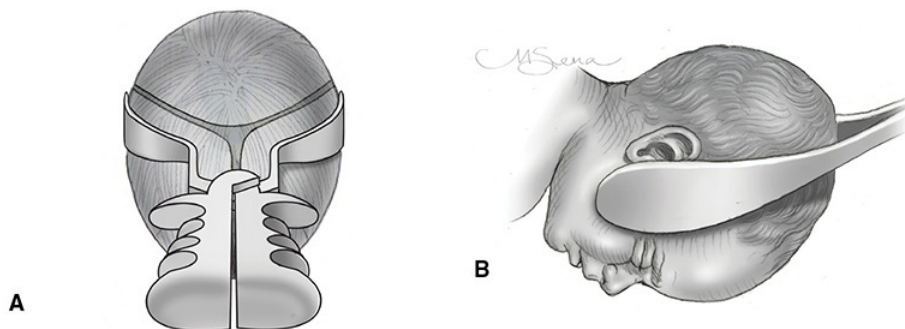


Figure 29-6. Correct symmetric application: blades along the sides of the head over the lateral face with the vertex OA.

Rotation and traction

- **Rotate to OA before traction.**
- Traction: **gentle, intermittent, downward and outward**, with maternal pushing, **only with contractions**. Initially downward (head under the symphysis), then **gradually upward** as the head descends, mimicking spontaneous birth.
- **Bill axis traction device** (teaching): arrow on the indicator line = path of least resistance.
- Force can't be measured → slow, deliberate, gentle (avoid head compression) unless urgent.
- As the occiput distends the vulva: episiotomy if needed. Either keep blades on (↑ distention/laceration), or **remove branches in the reverse order of placement** and finish with pushing ± modified Ritgen, **bolstering the perineum**. Removing too early → head recedes.

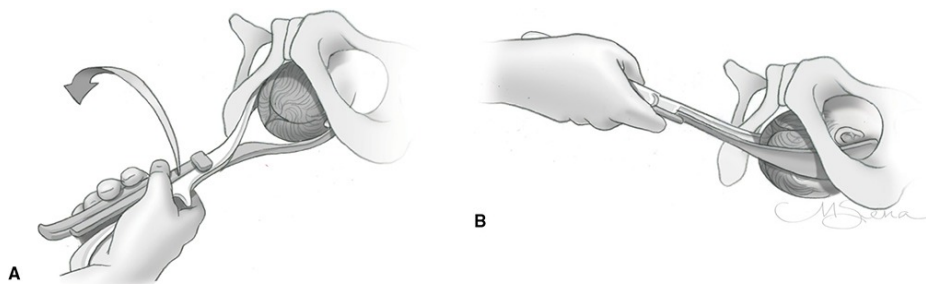


Figure 29-7. From LOA, the vertex is rotated to OA before traction.

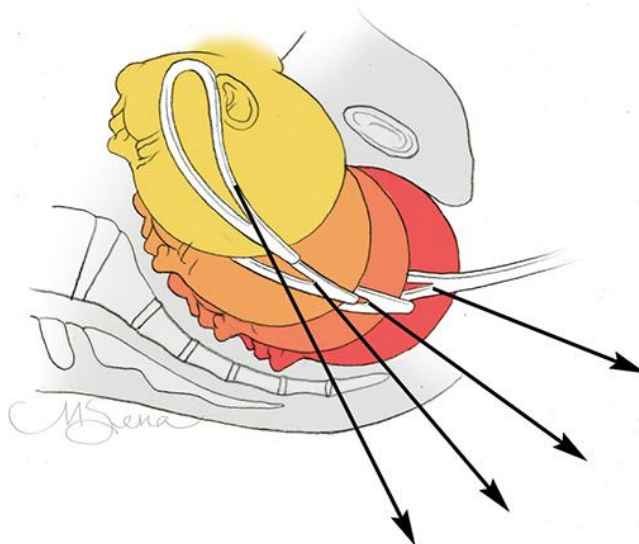


Figure 29-8. Low forceps: the traction vector moves from downward to upward as the head descends.

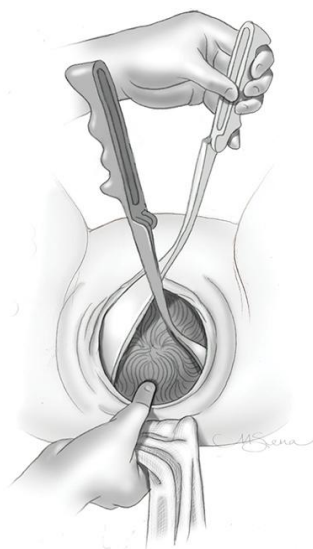


Figure 29-9. Branches removed in reverse order; a towel-covered hand bolsters the perineum and the thumb prevents sudden egress.

Manual rotation

- OT/OP heads are often **deflexed** (wider diameter) → slow descent. Rotating to OA improves proportions. Usually done in the **second stage**.
- vs no attempt: ↓ **cesarean**, ↓ **OVD**, ↓ **severe laceration**, ↓ **chorioamnionitis**. If OVD is still needed, OA ↓ perineal trauma.
- **Digital rotation**: two fingers hooked on the molded suture ridge, force parallel to the skull, **between contractions**; hold gains during pushing; repeat as needed.

- **Manual rotation** (open hand):
 - **ROP**: right palm cups the head over the sagittal suture, rotate **clockwise** → ROA.
 - **LOP**: left palm, **counterclockwise**.
 - **Three simultaneous actions** between contractions: **flex** the head, slightly **destation** it (⚠ **never disengage**), and use the other hand on the abdomen to pull the fetal back toward the midline.
- **Success >90%** (Le Ray, 796 rotations): no cord prolapse; 2 cervical lacerations; FHR normal in **71%**, severe FHR abnormalities in **10%**. Failure to rotate **does not mandate cesarean**.

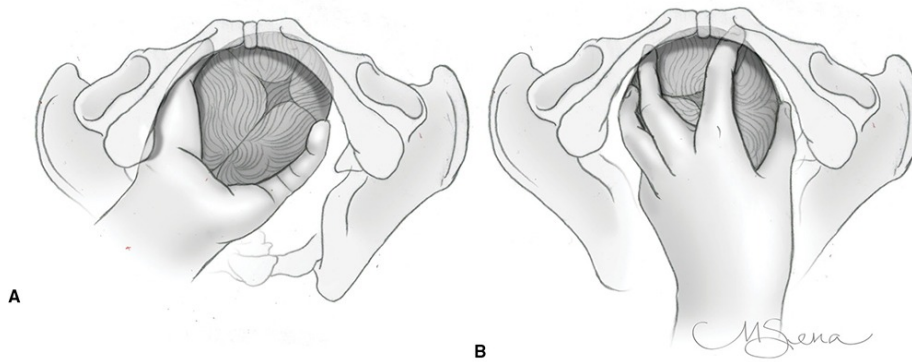


Figure 29-10. Manual rotation from ROP: the head is flexed and destationed during clockwise rotation to OA.

Note: the printed Fig 29-10 legend says the left hand rotates from ROP, whereas the text says the right palm is used for ROP (left palm for LOP).

Persistent occiput posterior

- Options: spontaneous OP delivery, manual/instrumental rotation to OA, or forceps/vacuum delivery **as OP**.
- Common cause: **anthropoid pelvis** (opposes rotation).
- **Forceps delivery from OP**: downward/outward traction until the **base of the nose passes under the symphysis** → **elevate handles** so the occiput emerges over the fourchette → then downward again for nose, mouth, chin.
- OP → more vulvar distention, **severe lacerations and extensive episiotomy**. Newborns delivered by OVD from OP: **Erb palsy 1%, facial palsy 2%**.
- **Kielland forceps** preferred for OP → OA rotation (**minimal pelvic curve, sliding lock, light weight**).

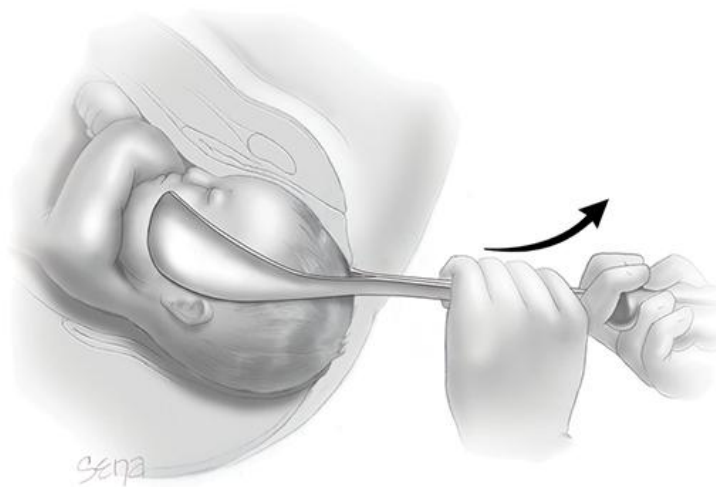


Figure 29-11. Outlet forceps from OP: the head should be flexed after the bregma passes under the symphysis.

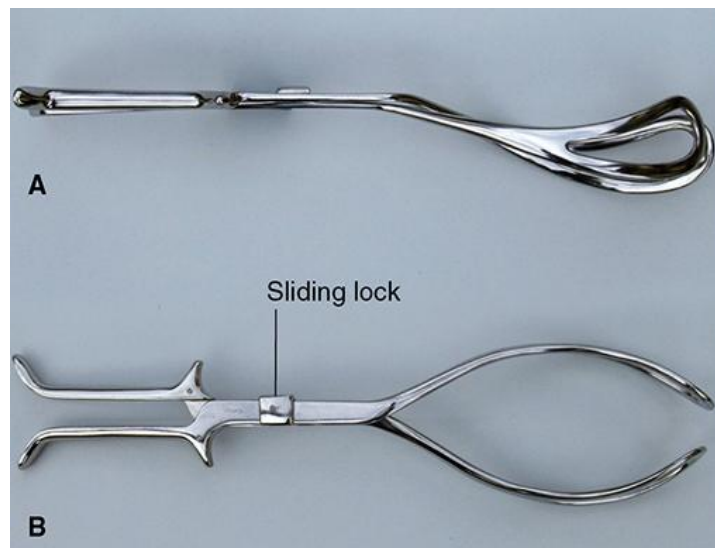


Figure 29-12. Kielland forceps: minimal pelvic curve, sliding lock, light weight.

Occiput transverse

- Rotation is required: manual, standard forceps or **Kielland** (high success, low morbidity in experienced hands).
- Kielland handles carry a **knob that faces the occiput**.
- **Two anterior-blade methods (LOT example):**
 1. **Wandering:** anterior blade introduced **posteriorly**, then arched around the face to the anterior position (handle kept near the left maternal buttock); second blade posteriorly along the sacral hollow; lock.
 2. **Classic:** anterior blade inserted with the **cephalic curve upward** under the symphysis, advanced, then **turned 180 degrees** on its long axis; second blade posterior; lock.
- **Rotation:** pull handles slightly to the patient's right to **flex**; fingers of the left hand on the finger guards; right-hand fingers on the **anterior lambdoid suture** (ensures the head turns with the blades); elevate/slightly destation; supinate the left wrist for **counterclockwise** rotation. Abdominal sonography can document rotation.
- **Delivery after rotation:**
 - Traction on the Kielland with a bimanual grip; once the posterior fontanel is under the subpubic arch, raise the handles **to horizontal but no further** (reverse pelvic curve → **vaginal sulcus tears**), or
 - Seat the head with moderate traction, then **swap to conventional forceps**.

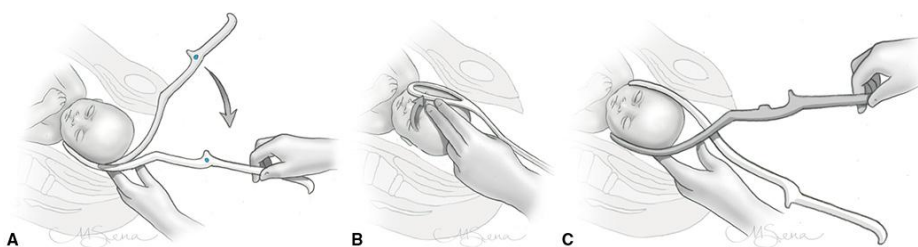


Figure 29-13. Kielland application for LOT: right (anterior) branch wandered into place behind the symphysis, then the left branch inserted posteriorly.

Face presentation

- **Mentum anterior:** forceps OK. Blades along the **occipitomenal** diameter, **pelvic curve toward the neck**; downward traction until the **chin is under the symphysis**, then upward: nose, eyes, brow, occiput.
- **⚠ Mentum posterior: no forceps** (vaginal delivery impossible except in very small fetuses). **Vacuum is not used for face presentation.**

5. Vacuum-Assisted Delivery

Design

- US: **vacuum extractor**; Europe: **ventouse**.
- **Theoretical advantages over forceps**: less need for precise placement; no space-occupying blades → less maternal trauma.
- Components: **cup, shaft, handle, vacuum generator** (handheld by operator or run by assistant).

Cup type	Traits
Soft bell cup (pliable dome)	Fewer scalp lacerations ; recommended for straightforward OA
Rigid mushroom cup (flat, circular rim ridge)	More traction force ; better at the flexion point in OP/asynclitism ; more scalp lacerations
Metal cup	Higher success but more scalp injury incl. cephalohematoma

- Cephalohematoma and subgaleal hemorrhage rates are **similar** for soft vs rigid cups (Vacca). **High pressure generates large force whatever the cup**.
- **Flexible shafts** help cup seating in OP/asynclitism.

Table 29-2. Vacuum Cups for Operative Vaginal Delivery

Cup Style	Manufacturer
Soft Bell Cup	
GentleVac	OB Scientific
Kiwi ProCup	Clinical Innovations
MitySoft	CooperSurgical
Pearl Edge	CooperSurgical
Mityvac Reusable Silicone ^a	CooperSurgical
Secure Cup	Utah Medical Products
Soft Touch	Utah Medical Products
Tender Touch	Utah Medical Products
Tender Touch Ultra	Utah Medical Products
Velvet Touch ^a	Utah Medical Products
Silc Cup	Medela
CaesAid ^a	Medela
Rigid Mushroom Cup	
Flex Cup	Utah Medical Products
M-Style	CooperSurgical
Super M-Style	CooperSurgical
M-Select ^b	CooperSurgical
Kiwi OmniCup ^b	Clinical Innovations
Kiwi Omni-MT	Clinical Innovations
Kiwi Omni-C ^c	Clinical Innovations
Bird Cup ^b	Medela
Malmström Cup	Medela

^aReusable cups. ^bSuitable for occiput posterior positions or asynclitism. ^cFor extractions through a hysterotomy incision during cesarean delivery.



Figure 29-14. Kiwi OmniCup (rigid mushroom cup, handheld pump) and Mityvac MitySoft Bell Cup (soft bell cup).

Technique

- **Flexion point:** on the **sagittal suture ~3 cm from the posterior fontanel and ~6 cm from the anterior fontanel**.
 - Cups are **5–6 cm** across → rim at the posterior fontanel border, **3 cm from the anterior fontanel**.
 - Benefits: max traction, fewer pop-offs, flexes the neck, **smallest diameter** delivered → ↑ success, ↓ scalp and perineal injury.
 - ⚠ **Anterior placement** (near the anterior fontanel) → neck extension, wider diameter. **Asymmetric** placement worsens asynclitism.
 - Easy in OA; hard when the indication is malposition-related failure to descend.
- **Check the whole cup rim** (before and after suction and before traction) for trapped **maternal soft tissue** (→ lacerations and "pop-off").
- **Vacuum build-up:** some raise it by **0.2 kg/cm² every 2 min** to **0.8 kg/cm²**; reaching 0.8 kg/cm² in **<2 min** is equally effective and safe.

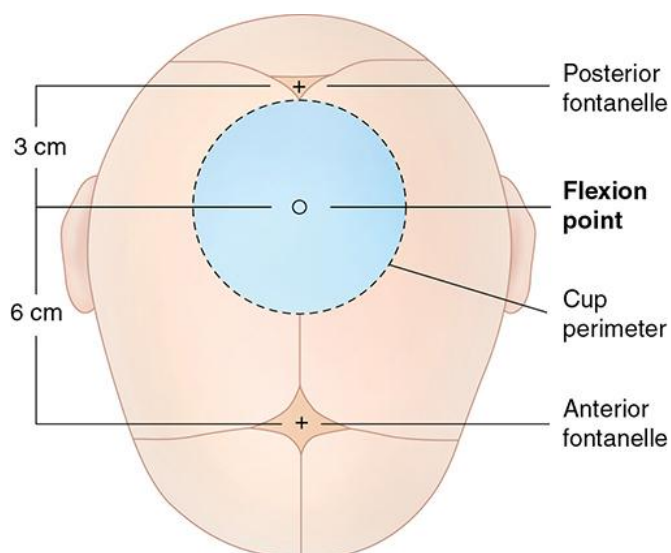


Figure 29-15. Flexion point: on the sagittal suture 3 cm from the posterior and 6 cm from the anterior fontanel.

Table 29-3. Vacuum Pressure Conversions

mm Hg	cm Hg	Inches Hg	lb/in ²	kg/cm ²
100	10	3.9	1.9	0.13
200	20	7.9	3.9	0.27
300	30	11.8	5.8	0.41
400	40	15.7	7.7	0.54
500	50	19.7	9.7	0.68
600	60	23.6	11.6	0.82

Memory hook: Target vacuum **0.8 kg/cm² ≈ 600 mm Hg**.

Traction

- Same angles as forceps (Fig 29-8); **steady, no jerking or rocking; 1–4 pulls** typical (more at higher station); intermittent, with pushing.
- **⚠ No manual torque/rotation** (cup displacement, cephalohematoma, "cookie-cutter" lacerations with metal cups). OA-oblique and OP heads are **not rotated**; with correct placement and traction line the head **rotates by itself**.
- **Nondominant hand in the vagina:** thumb on the cup, finger(s) on the scalp → judge descent, adjust the angle, detect cup separation.
- Lowering suction between contractions vs maintaining it: **no outcome difference**.
- Release vacuum once the head delivers.
- **It is a trial:** expect **progressive descent with each pull**. No consensus on maximum pulls, pop-offs or duration (manufacturers give guidance).
 - Pop-off from **technical failure** (poor placement) ≠ pop-off with ideal placement → may reapply or try forceps.
 - **Traction without progress or repeated detachment despite correct use** is the worst scenario → **be willing to abandon**.

6. Training

- Fewer OVDs → fewer training cases: median **~20** for recent graduates; ACGME **minimum 15** (2021).
- Solutions: skilled teachers, simulation.
 - One dedicated instructor → **59% ↑** forceps deliveries over 2 years.
 - Manikin/pelvic simulation program → ↓ neonatal morbidity and severe cervical/labial/vaginal lacerations (OASIS unchanged).
 - OVD simulation curriculum → **22% ↓ OASIS**.

Quick-Recall: Forceps vs Vacuum

Feature	Forceps	Vacuum
Rotation	Yes (manual rotation, Kielland)	No (traction only; head self-rotates)
Face (mentum anterior)	Yes	No
Aftercoming breech head	Yes (Piper)	No
Minimum GA	—	≥34 wk
Failure (all US)	0.4%	0.8%
Failure (term nulliparas)	4.4%	6.4%
Maternal injury (OASIS)	Higher	Lower
Fetal injury profile	Facial nerve, brachial plexus, depressed skull fracture, corneal abrasion	Cephalohematoma, subgaleal, retinal hemorrhage , jaundice, scalp laceration
Total neuro complications (Werner)	Fewer	More
Key placement landmark	Blades along occipitomental diameter, equidistant from sagittal suture	Flexion point : 3 cm from posterior, 6 cm from anterior fontanel
Rotational OVD vs second-stage cesarean	Similar morbidity	Similar morbidity

Key Numbers to Memorize

- OVD **3.1%** of US births (was **9%** in 1990); vacuum:forceps **4:1**
- Station **-5 to +5**; outlet = on the perineum, rotation **≤45°**; low = **≥+2** not on floor; mid = **0 to +2**; high = above 0 (**never**)
- Vacuum not before **34 wk**; retention resolves with **24–48 h** catheter drainage
- ANODE: infection **11% vs 19%** with single-dose amoxicillin–clavulanate
- OP OVD: Erb palsy **1%**, facial palsy **2%**
- Manual rotation success **>90%**; severe FHR changes **10%**
- Fetal head: occipitofrontal **11.5**, occipitomental **12.5**, suboccipitobregmatic **9.5**, biparietal **9.5 cm**
- Flexion point **3 cm / 6 cm**; cup **5–6 cm**; vacuum **0.8 kg/cm² (≈600 mm Hg)**; step-up **0.2 kg/cm² per 2 min**; **1–4 pulls**
- Failure predictors: **OP**, birthweight **>4000 g**
- ACGME minimum **15** OVDs; simulation → **22%** ↓ OASIS